SECTION 0: PRE-ANALYSIS REFLEXIVE DECLARATION

Likely home grammar

My likely “home grammar” for antimicrobial resistance (AMR) analysis is a **systems public-health governance grammar**. This framework typically treats AMR as a multi-sector collective-action problem requiring coordinated policy across medicine, agriculture, and pharmaceutical innovation. It prioritizes epidemiological projections, stewardship programmes, and economic modelling of intervention costs.

Hardest grammar to operate

The most difficult grammar for me to operate faithfully is likely the **market-innovation grammar** that frames AMR primarily as a pharmaceutical R&D incentive failure. The challenge arises because this grammar treats microbial evolution and antibiotic misuse as secondary to investment structures, whereas public-health approaches tend to view stewardship and surveillance as central.

Potential blind spots

Given that orientation, I may under-recognize arguments that:

- Regulatory and stewardship systems suppress antibiotic innovation.
- Agricultural productivity imperatives legitimately justify certain antibiotic uses.
- Market dynamics—not governance—drive long-term antibiotic availability.

I may also implicitly assume the One-Health paradigm is normatively correct rather than itself an interpretive construction.

Temporal orientation

My analytical tendency is **forward-prescriptive**: evaluating signals primarily in terms of future risk management and policy design rather than purely historical explanation or interpretive meaning-making.

SECTION 1: OBSERVATION FIELD DECLARATION

Phenomenon

The case concerns **global responses to antimicrobial resistance (AMR)**—the increasing ability of bacteria, fungi, parasites, and viruses to survive treatments previously effective against them.

Temporal Frame

2015–2026, beginning with the **WHO Global Action Plan on AMR (2015)** and continuing through present global policy initiatives, surveillance expansions, and pharmaceutical investment debates.

Geographic Scope

Global, with specific attention to:

- High-income countries (US, EU, UK, Japan)
- Rapidly industrializing regions (China, India, Brazil)
- Low-income regions with limited surveillance capacity (sub-Saharan Africa, parts of South Asia)

Why This Domain Produces Interpretive Divergence

At first glance AMR appears consensual: virtually all actors agree that antimicrobial resistance is dangerous.

However, divergence arises because:

- **Causal interpretation differs**: misuse vs innovation failure vs ecological inevitability.
- **Policy priorities conflict**: stewardship vs drug development vs agricultural productivity.
- **Economic incentives diverge** between governments, hospitals, farmers, and pharmaceutical companies.
- **Surveillance inequality** makes empirical signals uneven across regions.

Thus the same microbiological and epidemiological evidence can produce radically different conclusions.

Signal Contestation Level

Mixed domain:

- Many signals are **shared** (e.g., mortality estimates, antibiotic consumption).
- Several are **contested** (e.g., agricultural contribution to resistance, pipeline sufficiency).

Major Interpretive Communities

- Global health institutions (WHO, CDC, ECDC)
- Pharmaceutical industry and biotech investors
- Agricultural and livestock sectors
- Health economists
- Development policy actors

- National governments
- Microbiologists and infectious-disease clinicians

SECTION 2: SIGNAL SET

Core Signals

1. **Global AMR mortality estimate (2019)**
   Source: Lancet Global Burden of Disease AMR study.
   Type: Epidemiological estimate.
   Data: ~1.27 million deaths directly attributable; ~4.95 million associated.
   Ambiguity: Attribution modelling depends on assumptions about counterfactual treatments.

2. **Global antibiotic consumption growth (2000–2018)**
   Source: IQVIA / global surveillance analyses.
   Type: Usage trend.
   Observation: Large increase in LMICs, modest stabilization in some high-income countries.
   Ambiguity: Growth may reflect improved access rather than misuse.

3. **Agricultural antibiotic usage**

Source: FAO/OIE/various national datasets.
Type: Sectoral consumption data.
Observation: Large livestock usage in China, US, Brazil, India.
Ambiguity: Data incomplete; prophylactic vs therapeutic use often blurred.

4. **Antibiotic pipeline**

Source: WHO antibiotic pipeline reports (2021–2024 updates).
Type: Innovation indicator.
Observation: Limited number of truly novel classes; many derivatives.
Ambiguity: “Novelty” definitions vary.

5. **Market withdrawals and bankruptcies**

Source: industry reports (Achaogen, Melinta, others).
Type: economic signal.
Observation: Several antibiotic startups went bankrupt after approval due to low sales.
Ambiguity: Could indicate stewardship success rather than market failure.

6. **Stewardship programme adoption**

Source: OECD/WHO surveys.
Type: policy implementation metric.

Observation: Stewardship programs widespread in high-income hospitals but uneven globally.
Ambiguity: Implementation quality varies widely.

7. **One-Health framework adoption**

Source: WHO/FAO/OIE initiatives.
Type: governance framework.
Observation: Increasing policy adoption linking human, animal, and environmental health.
Ambiguity: Often symbolic; operationalization unclear.

8. **Surveillance infrastructure (GLASS)**

Source: WHO Global Antimicrobial Resistance Surveillance System.
Type: data capacity indicator.
Observation: Expanding participation but uneven reporting quality.
Ambiguity: Low-income countries have significant data gaps.

9. **Economic cost projections**

Source: UK O'Neill Review (2016), OECD models.
Type: economic forecast.
Projection: Up to 10 million deaths annually by 2050 under worst scenarios.
Ambiguity: Model assumptions contested.

10. **Decline in antibiotic discovery by major pharma**

Source: industry R&D data.
Type: investment trend.
Observation: Large pharmaceutical firms have largely exited antibiotic development.
Ambiguity: Could reflect scientific difficulty rather than economic structure.

11. **Policy instruments for antibiotic incentives**

Source: US PASTEUR Act proposals, EU subscription models, UK pilot programme.
Type: policy experiment.
Observation: "Delinked" payment models emerging.
Ambiguity: Long-term effectiveness unknown.

12. **National bans on growth-promotion antibiotics**

Source: EU ban (2006), China restrictions (2020), others.
Type: regulatory change.
Ambiguity: Enforcement inconsistent.

13. **Regional resistance patterns**

Source: ECDC, CDC surveillance.
Observation: Carbapenem-resistant Enterobacteriaceae increasing in some regions.
Ambiguity: Data quality varies.

Counter-Signals

1. **Improved access argument**

Rising antibiotic use in LMICs may reflect improved treatment access rather than overuse.

2. **Agriculture impact uncertainty**

Some studies find limited direct transfer from livestock antibiotic use to human clinical resistance.

3. **Pipeline may be underestimated**

Biotech innovation, phage therapy, and non-traditional approaches may not appear in traditional pipeline metrics.

4. **Mortality attribution uncertainty**

AMR death estimates depend on modelling assumptions.

5. **Local success stories**

Some countries (e.g., Netherlands, Denmark) have dramatically reduced livestock antibiotic use without harming productivity.

Contested Signals

1. **Agricultural antibiotic contribution**

- Public-health grammar: major driver of resistance evolution.
- Market-innovation grammar: secondary contributor relative to clinical misuse.
- Food-production grammar: minimal or poorly evidenced impact.

2. **Pipeline inadequacy**

- Innovation grammar: catastrophic underinvestment.
- Public-health grammar: moderate problem but stewardship more important.

- Agricultural grammar: largely irrelevant to livestock systems.

3. **Mortality projections**

- Some grammars treat projections as policy-relevant warnings; others see them as speculative modelling.

SECTION 3: GRAMMAR SPECIFICATIONS

Three grammars dominate AMR interpretation.

Grammar A: Global Health Security Grammar

Primary analytical variable
Population-level infection risk.

Signal selection rules
Admits epidemiological data, resistance surveillance, antibiotic consumption, and mortality estimates.

Weighting priority
Public-health outcomes (mortality, resistance prevalence).

Causal chain type
Misuse → selection pressure → resistance emergence → treatment failure → global health threat.

Admissible closure
Strong stewardship, surveillance expansion, reduced antibiotic use, One-Health coordination.

Excluded signals
Pharmaceutical profitability concerns unless they affect access.

Characteristic vocabulary
“Global health threat,” “stewardship,” “surveillance,” “One Health.”

Temporal orientation
Forward-prescriptive.

Source tradition
WHO, CDC, global health epidemiology.

Grammar boundary
If pharmaceutical innovation becomes the central explanatory variable, the grammar shifts into the **Market Innovation Grammar**.

Grammar B: Market Innovation Grammar

Primary analytical variable
Economic incentives for antibiotic R&D.

Signal selection rules
Admits pharmaceutical investment trends, pipeline metrics, bankruptcies, and policy incentives.

Weighting priority
Drug discovery capacity.

Causal chain type
Weak incentives → insufficient antibiotic development → shrinking treatment arsenal → future crisis.

Admissible closure
Market reforms (subscription models, prize funds, delinked payments).

Excluded signals
Agricultural use unless it affects demand.

Characteristic vocabulary
“Innovation ecosystem,” “market failure,” “delinkage.”

Temporal orientation
Forward-prescriptive but economically framed.

Source tradition
Health economics, pharmaceutical policy.

Grammar boundary
If antibiotic misuse becomes the core causal driver, the grammar transforms into the **Global Health Security Grammar**.

Grammar C: Food Production & Development Grammar

Primary analytical variable
Food security and agricultural productivity.

Signal selection rules
Admits livestock antibiotic use, agricultural yield data, rural economic impacts.

Weighting priority
Food production stability.

Causal chain type
Livestock disease risk → antibiotic use → stable food supply.

Admissible closure
Targeted reduction of non-therapeutic use but preservation of veterinary antibiotics.

Excluded signals
Most pharmaceutical pipeline data.

Characteristic vocabulary
“Food security,” “livestock health,” “production efficiency.”

Temporal orientation
Present-pragmatic.

Source tradition
Agricultural economics, livestock industry.

Grammar boundary
If resistance spillover to humans becomes the central causal variable, this grammar shifts toward the Global Health Security grammar.

SECTION 4: SIGNAL PROCESSING

Grammar A Processing: Global Health Security

The Global Health Security grammar begins with **Signal 1: AMR mortality estimates**. The figure of 1.27 million deaths directly attributable to AMR establishes the phenomenon as a large-scale public health threat. Even acknowledging modelling ambiguity (counter-signal: attribution uncertainty), the magnitude positions AMR alongside diseases such as HIV or malaria.

Next, the grammar processes **Signal 2: rising global antibiotic consumption**. From this perspective, increasing antibiotic use—especially in rapidly developing regions—represents increased **selection pressure on microbial populations**. Even where increased use reflects improved access (counter-signal), the evolutionary mechanism remains the same: every exposure event creates opportunity for resistance selection.

Signal 13: regional resistance patterns reinforces the narrative. Carbapenem-resistant pathogens, once rare, are increasingly observed in hospitals worldwide. This signal fits the grammar's causal chain: antibiotic exposure leads to resistant strains spreading through healthcare systems.

The grammar then integrates **Signal 3: agricultural antibiotic use**. Within this interpretive architecture, livestock use constitutes an additional reservoir of selection pressure. Even though the precise magnitude of spillover to human pathogens is debated (counter-signal), the One-Health framework (Signal 7) encourages precaution: microbial ecosystems connect human, animal, and environmental compartments.

Signal 12: national bans on growth-promotion antibiotics is interpreted as partial validation of this framework. Countries implementing these restrictions represent policy attempts to reduce non-essential antibiotic use. However, inconsistent enforcement across regions demonstrates incomplete adoption.

Signal 6: stewardship programme adoption becomes a central intervention signal. In high-income healthcare systems, antimicrobial stewardship programs aim to optimize prescribing practices. The grammar reads the uneven global adoption as a major vulnerability: resistance evolves fastest where governance capacity is weakest.

Signal 8: surveillance infrastructure (GLASS) is critical because without reliable data, resistance patterns cannot be tracked. In regions lacking laboratory infrastructure, resistant pathogens may circulate undetected. This signal reveals a structural asymmetry: the places where AMR may expand fastest are often the places with the least surveillance capacity.

Economic signals such as **Signal 9: AMR cost projections** reinforce urgency. Models projecting millions of deaths by 2050 are interpreted less as precise forecasts and more as **policy warning indicators**. Even if uncertain, the scale justifies precautionary policy.

When encountering **Signal 4: the weak antibiotic pipeline**, the grammar acknowledges risk but treats it as secondary. Stewardship measures aim to preserve existing drugs; therefore innovation, while important, does not replace the need to reduce misuse.

Counter-signals introduce tension. For example, the **improved access argument** suggests antibiotic consumption increases in LMICs represent progress in healthcare access rather than misuse. The grammar partially accepts this point but maintains that **access and stewardship must expand simultaneously**.

Similarly, **agricultural impact uncertainty** challenges the assumption that livestock antibiotic use significantly affects human health. Within this grammar the response is precautionary: given microbial gene transfer across ecosystems, even uncertain pathways justify reduction efforts.

A counterpoint exists inside the grammar: overly aggressive stewardship could reduce antibiotic access for patients in low-resource settings. Infectious disease physicians sometimes worry that restrictive policies delay treatment.

Despite this tension, the grammar reaches closure:

- AMR represents a **global health security threat driven primarily by antibiotic overuse and insufficient surveillance**.
- The appropriate response includes stewardship expansion, surveillance investment, agricultural regulation, and coordinated One-Health governance.

This closure emerges directly from the grammar's emphasis on **population-level infection risk** and evolutionary selection pressure.

The conclusion should therefore be understood as a product of the **Global Health Security processing architecture** rather than an inevitable reading of the evidence.

Grammar B Processing: Market Innovation

The Market Innovation grammar begins from a different signal: **Signal 10: pharmaceutical exit from antibiotic R&D**. Large pharmaceutical firms have largely abandoned antibiotic development over the past two decades. This signal is interpreted as the foundational causal problem.

Next, **Signal 5: antibiotic company bankruptcies** strengthens the diagnosis. Companies such as Achaogen developed clinically valuable antibiotics but failed commercially because stewardship policies limited sales volumes. From this grammar's perspective, the failure is structural: antibiotics are intentionally used sparingly, making them economically unattractive products.

Signal 4: the antibiotic pipeline becomes the central indicator of system health. WHO reports show relatively few truly novel antibiotic classes in development. For the innovation grammar, this suggests that **future treatment options are collapsing**.

The grammar processes **Signal 1: AMR mortality estimates** differently than the health-security grammar. Mortality statistics are not primarily evidence of misuse but evidence that **current treatments are losing effectiveness faster than replacements appear**.

Signal 9: economic cost projections reinforce this interpretation. Models predicting catastrophic economic impact are seen as confirmation that the antibiotic innovation ecosystem is failing.

Policy experiments such as **Signal 11: subscription payment models** become central solutions. Under these systems, governments pay pharmaceutical firms for guaranteed antibiotic availability rather than per-dose sales, decoupling revenue from usage volume.

When encountering **Signal 6: stewardship programmes**, the grammar views them ambivalently. Stewardship is necessary to preserve drug effectiveness, but it also suppresses demand, worsening the investment environment.

Signal 2: rising antibiotic consumption produces mixed interpretation. On one hand, increased use might signal misuse and resistance pressure. On the other, it indicates that antibiotics remain essential medical tools with large demand. The key issue is not consumption itself but the absence of **innovative replacements**.

Signal 3: agricultural antibiotic use is treated as peripheral. Even if agriculture contributes somewhat to resistance, this grammar argues that **the decisive variable is therapeutic capability**: if new antibiotics exist, resistant strains can still be treated.

Counter-signals introduce challenges. For example, **local stewardship successes** in Denmark or the Netherlands show that agricultural antibiotic use can fall dramatically without harming productivity. This suggests that misuse reduction may be achievable without sacrificing food security. However, the innovation grammar interprets these examples as **demand suppression that further discourages R&D investment**.

Another counter-signal is the emergence of **non-traditional therapies** such as phage therapy or monoclonal antibodies. These developments complicate the narrative of innovation collapse, though many remain early stage.

A counterpoint within the grammar acknowledges that **incentive reform alone cannot prevent resistance evolution**. Microbes inevitably adapt; therefore stewardship remains necessary.

Despite this, the grammar's closure is clear:

- The AMR crisis is fundamentally an **innovation market failure**.
- Without restructuring antibiotic incentives, the world will run out of effective drugs regardless of stewardship efforts.

Policy prescriptions include subscription models, large innovation prizes, public-private partnerships, and regulatory reforms to stabilize antibiotic markets.

This closure emerges from the grammar's emphasis on **economic incentives for drug discovery**.

Grammar C Processing: Food Production and Development

The Food Production grammar begins with **Signal 3: agricultural antibiotic use** but interprets it differently from the health-security grammar. Here antibiotics function primarily as **tools for maintaining livestock health and production efficiency**.

Livestock production systems often involve dense animal populations vulnerable to infectious disease outbreaks. Antibiotics reduce mortality and maintain stable food supply chains. Therefore the baseline interpretation of agricultural antibiotic use is **functional necessity**, not misuse.

Next, the grammar processes **Signal 12: national bans on growth-promotion antibiotics**. These bans represent regulatory attempts to reduce non-therapeutic use. The grammar acknowledges them but emphasizes that bans must avoid undermining livestock productivity.

Signal 2: global antibiotic consumption trends are interpreted partly as development indicators. Rising consumption in low-income regions reflects expanding access to both human and veterinary medicine, which may support agricultural modernization.

When encountering **Signal 1: AMR mortality estimates**, the grammar recognizes the seriousness of resistant infections but does not automatically assign major causal responsibility to agriculture. The key issue becomes **how much resistance emerges from human medicine versus livestock environments**.

This is where **Contested Signal 1** becomes central. Studies demonstrating genetic links between livestock pathogens and human infections exist, but the magnitude of spillover remains debated. Within this grammar, the uncertainty reduces the urgency of drastic agricultural restrictions.

The **One-Health framework (Signal 7)** is processed cautiously. While conceptually appealing, it may lead to regulatory burdens on farmers without clear evidence of benefit.

Signal 6: stewardship programmes in human hospitals are viewed positively because they reduce unnecessary prescriptions without affecting food production.

The grammar largely excludes **Signal 10 (pharmaceutical investment trends)** and **Signal 5 (biotech bankruptcies)** because they do not directly affect livestock disease management.

Counter-signals challenge this interpretation. For example, **successful livestock antibiotic reductions in Denmark and the Netherlands** demonstrate that production can remain stable even after substantial reductions. These cases introduce pressure on the grammar: if productivity can be maintained with less antibiotic use, the justification for widespread prophylactic use weakens.

Another counter-signal is emerging genomic evidence linking certain resistance genes in humans to agricultural sources. While still debated, such findings create potential tension.

A counterpoint within the grammar recognizes that **uncontrolled resistance could eventually threaten veterinary medicine itself**. If antibiotics lose effectiveness, livestock disease outbreaks could become harder to control.

Despite these tensions, the grammar's closure remains pragmatic:

- Agricultural antibiotic use should be **reduced where unnecessary**, particularly growth promotion.
- However, veterinary antibiotics remain essential for maintaining global food supply and rural livelihoods.
- Policy should focus on **improved husbandry, targeted treatment, and gradual reduction rather than aggressive bans**.

This closure arises from the grammar's primary commitment to **food security and agricultural stability**.

SECTION 5: CLOSURE COMPARISON

Closure Content

Each grammar produces a distinct policy diagnosis.

- **Global Health Security grammar:** AMR arises primarily from antibiotic misuse and weak surveillance. Solution: stewardship and One-Health governance.
- **Market Innovation grammar:** AMR results from a broken pharmaceutical innovation ecosystem. Solution: economic incentives for new antibiotics.
- **Food Production grammar:** AMR policies must balance resistance risk with food security. Solution: cautious reduction of agricultural antibiotics without harming productivity.

All three grammars accept that resistance exists but diverge sharply on **what the core problem actually is**.

Signal Divergence

Different grammars elevate different signals:

- Health security emphasizes **mortality estimates, resistance surveillance, and antibiotic consumption trends**.
- Market innovation emphasizes **pipeline data, pharmaceutical exits, and investment structures**.
- Food production emphasizes **livestock antibiotic usage and agricultural productivity metrics**.

Shared signals therefore become **interpretively reweighted rather than disputed outright**.

Exclusion Patterns

Each grammar suppresses certain signals:

- Health security downplays **pharmaceutical profitability**.
- Market innovation marginalizes **agricultural usage debates**.
- Food production largely ignores **pharmaceutical pipeline metrics**.

These exclusions significantly shape each grammar's closure.

Causal Architecture

Causal chains differ fundamentally:

Health Security
misuse → resistance evolution → treatment failure

Market Innovation
weak incentives → innovation collapse → treatment scarcity

Food Production
livestock disease risk → antibiotic use → food supply stability

Thus the same phenomenon—antibiotic resistance—is embedded in different causal systems.

Closure Range

The admissible conclusions differ:

- Health security allows **strong regulatory intervention**.
- Market innovation allows **large economic subsidies to pharmaceutical companies**.
- Food production allows **incremental agricultural reform** but rejects sweeping bans.

Closure Timing

The grammars reach closure at different speeds.

- Health security reaches closure quickly once mortality and resistance signals are processed.
- Market innovation requires processing multiple economic signals before closure.
- Food production reaches closure rapidly because it begins with a normative commitment to food security.

Ontological Check

The grammars partially **constitute different objects**.

- Health security analyzes **AMR as a population-level epidemiological threat**.
- Market innovation analyzes **AMR as a pharmaceutical innovation system failure**.
- Food production analyzes **AMR policy as an agricultural management problem**.

Thus they are not always analyzing the same phenomenon; they analyze **different systems linked by the term "AMR."**

SECTION 6: COUNTER-SIGNALS, LEAKAGE, AND STRESS TEST

Consolidated Counter-Signals

Major counter-signals affecting all grammars include:

- Uncertainty in AMR mortality estimates.
- Ambiguity regarding agricultural contribution to human resistance.
- Evidence that some countries successfully reduce antibiotic use without harming productivity.
- Emerging alternative therapies that may expand treatment options.

Leakage

Leakage occurs when grammars cannot fully suppress conflicting signals.

Examples:

- The **health security grammar** must acknowledge pharmaceutical innovation concerns because stewardship can reduce market incentives.
- The **market innovation grammar** cannot ignore stewardship entirely because resistance evolution would undermine new antibiotics.
- The **food production grammar** cannot ignore human health outcomes if resistance spillover evidence strengthens.

Grammar Stress Test

Global Health Security grammar
Most threatening counter-signal: evidence that agricultural antibiotic use contributes minimally to human AMR.
Absorption capacity: moderate. The grammar could still focus on human misuse and surveillance.

Market Innovation grammar
Most threatening counter-signal: strong evidence that stewardship and misuse reduction alone can control resistance without major new drugs.
Absorption capacity: weak. This would undermine the argument that innovation incentives are the central problem.

Food Production grammar
Most threatening counter-signal: definitive genomic evidence linking livestock antibiotic use to widespread

human resistance.
Absorption capacity: limited; it would require major regulatory changes in agriculture.

SECTION 7: POST-ANALYSIS REFLEXIVE NOTE

My predicted home grammar—**global health security**—was indeed the most natural framework for processing the AMR evidence. Mortality estimates, resistance prevalence, and antibiotic consumption trends are highly salient signals in global health discourse, making it easy to construct a coherent narrative centered on misuse and surveillance gaps.

The most difficult grammar to operate faithfully was **food production and development**. The difficulty arises because many public-health analyses implicitly treat agricultural antibiotic use as a major driver of resistance. To process signals through the food-production grammar required temporarily suspending that assumption and foregrounding food security concerns.

Interestingly, the **market innovation grammar** was easier to inhabit than expected. The evidence of repeated antibiotic startup bankruptcies and pharmaceutical exit from the field provides strong signals that can anchor a coherent narrative. Even if one does not accept this explanation as primary, the signals themselves are difficult to ignore.

A blind spot that emerged during the exercise concerns **ontological divergence**. Initially, it seemed that all grammars were analyzing the same phenomenon—antimicrobial resistance. However, the analysis revealed that each grammar effectively studies a **different system**:

- an epidemiological threat,
- a pharmaceutical market,
- or an agricultural production tool.

Recognizing this ontological divergence helps explain why debates about AMR policy often appear intractable. Participants are not merely disagreeing about solutions; they are analyzing **different causal systems** that share overlapping evidence.

If the analysis were conducted from a market-innovation home grammar, the signal set might have been constructed differently. Pipeline data, venture capital flows, and regulatory approval timelines would likely appear earlier and receive greater weight.

The exercise highlights a broader methodological insight: interpretive grammars shape not only conclusions but also **the structure of the problem itself**.

SECTION 8: METHODOLOGICAL OBSERVATIONS

The v2.0 version of the Prismatic Method worked particularly well in three areas.

First, the requirement to **integrate counter-signals during processing** significantly improved analytical realism. Rather than presenting grammars as perfectly coherent systems, the analysis revealed how each grammar actively manages internal tensions.

Second, the addition of **grammar boundaries** clarified the relationships between frameworks. In this case, the boundary between the health-security grammar and the innovation grammar is particularly thin: shifting the primary causal variable from misuse to R&D incentives effectively transforms one grammar into the other.

Third, the **ontological check** proved highly valuable. It revealed that the grammars sometimes construct different phenomena rather than merely different interpretations of the same object.

Some difficulties also emerged.

The distinction between **excluded signals and low-weighted signals** can be ambiguous. For instance, the health-security grammar does not fully exclude pharmaceutical investment signals; it simply treats them as secondary.

Another challenge concerns **signal contestation**. Determining whether a signal is truly shared or fundamentally grammar-dependent can require deep domain expertise.

The **stress-test stage** was useful because it exposed the conditional nature of some closures. The market-innovation grammar, in particular, appears highly sensitive to evidence that stewardship alone could stabilize resistance levels.

Possible refinements for future versions include:

- A more formal method for identifying **signal weighting** (perhaps scoring).
- Explicit mapping of **institutional actors** associated with each grammar.
- A step for analysing how grammars interact in real policy negotiations.

Overall, the method effectively demonstrates how a stable empirical field—microbial resistance data, antibiotic usage statistics, and economic models—can generate divergent policy conclusions depending on the interpretive grammar applied.